Bio Tech 9 (Biology Tech 9) — Solved Exercises (PCTB)

Class 9 — Solved Exercises

Punjab Curriculum and Textbook Board (PCTB)

9 chapters — 172 exercises

Chapter 1: The Science of Biology

A. Multiple Choice Questions

1. Heart pumps blood, stomach digests food, and kidneys excrete wastes. The statement comes from.

(a) Physiology ✓

(b) Anatomy

(c) Morphology

(d) Histology

Correct answer: Physiology

A. Multiple Choice Questions

2. Which step comes between making hypothesis and doing experiments?

(a) Making deductions ✓

(b) Making observations

(c) Summarizing results

(d) Analysing data

Correct answer: Making deductions

A. Multiple Choice Questions

3. Which of the following is NOT a characteristic of the scientific method?

(a) It relies on evidence

(b) It involves formulating hypotheses

(c) Hypothesis will always be correct ✓

(d) It requires rigorous testing

Correct answer: Hypothesis will always be correct

A. Multiple Choice Questions

4. Choose the correct sequence of steps of scientific method?

(a) Observations - hypothesis - deduction - experiments ✓

(b) Observations - hypothesis - law - theory

(c) Hypothesis - observations - deduction - experiments

(d) Law - theory - deduction - observations

Correct answer: Observations - hypothesis - deduction - experiments

A. Multiple Choice Questions

5. People who slept near smoky fire had less chance to suffer from malaria. Why?

(a) Smoke kills Plasmodium in their blood

(b) Fire increases temperature and Plasmodium are killed in air

(c) Mosquitoes cannot tolerate smoke and are repelled ✓

(d) Smoke kills Plasmodium present in mosquitoes

Correct answer: Mosquitoes cannot tolerate smoke and are repelled

A. Multiple Choice Questions

6. Experiments are very important in scientific method because a researcher:

(a) Always gets correct results

(b) Disproves many hypotheses and gets some hypothesis proved ✓

(c) Is sure that he will prove the hypotheses

(d) Gets a chance to work in the laboratory

Correct answer: Disproves many hypotheses and gets some hypothesis proved

B. Short Answers

1. Define the following branches of Biology: Genetics, Anatomy, Palaeontology, Marine Biology, Pathology.

Genetics is the study of heredity and variation, that is, how characters pass from parents to offspring. Anatomy is the study of the internal structure of organisms. Palaeontology is the study of fossils and ancient life. Marine Biology is the study

of organisms living in the sea. Pathology is the study of diseases, their causes and effects on the body.

B. Short Answers

2. Differentiate between Morphology and Physiology.

Morphology is the study of the form and external structure of organisms, such as the shape of leaves or the body plan of an animal. Physiology is the study of the functions of living things, such as how the heart pumps blood or how the stomach digests food. In short, morphology describes what an organism looks like, while physiology explains how it works.

B. Short Answers

3. What is Computational Biology?

Computational Biology is the branch of biology in which computers, mathematics and statistics are used to study biological data. It is used to analyse large sets of data such as DNA sequences, to model living processes and to predict how biological systems behave. It helps biologists handle data too large to study by hand.

B. Short Answers

4. What is the role of observation and experimentation in the scientific method?

Observation is the starting point: a scientist notices a problem or a pattern in nature and gathers facts about it, and from these observations a hypothesis is formed. Experimentation tests the hypothesis under controlled conditions, so that deductions drawn from it can be checked. If the results agree with the deductions, the hypothesis is supported; if not, it is rejected or changed.

C. Answers in Detail

1. Explain how the study of Biology can lead to different professional studies.

Biology is the foundation of many professions. Medicine, dentistry, nursing and pharmacy all rest on knowledge of the human body, diseases and drugs. Agriculture, horticulture and veterinary science apply biology to crops and animals. Other careers include biotechnology, food technology, fisheries, forestry and environmental science, where biological knowledge is used to improve health, food production and the environment.

C. Answers in Detail

2. How a hypothesis is converted to theory, law and principle?

A hypothesis is a testable statement proposed to explain observations. When it is repeatedly tested by experiments and supported by evidence, it may become a theory, which is a well-tested explanation of a broad set of facts. A law or principle is a statement that describes a regular pattern in nature, such as what always happens under given conditions. A theory explains why things happen, while a law describes what happens; neither is simply a guess.

C. Answers in Detail

3. What are the basic steps a scientist adopts in order to solve a scientific problem?

The scientist first makes observations and identifies a problem, then collects information about it. Next a hypothesis is proposed to explain the problem, and deductions are drawn from the hypothesis, that is, what should happen if it is true. Experiments are then performed to test these deductions, results are analysed, and a conclusion is drawn. If the results support the hypothesis it is accepted, otherwise it is modified and tested again.

C. Answers in Detail

4. Describe the work of different scientists in discovering the cause of malaria.

Laveran (1880) first found the parasite Plasmodium in the blood of malaria patients. Manson suggested that mosquitoes carry the disease from person to person. Ronald Ross then showed, by experiments on birds and mosquitoes, that the parasite develops inside the mosquito and is passed on when the mosquito bites. Later workers, including Grassi, confirmed that the female Anopheles mosquito transmits human malaria.

C. Answers in Detail

5. Write a descriptive note on the experiments performed by Ross.

Ross studied malaria in birds, which is similar to human malaria. He allowed mosquitoes to feed on malaria-infected birds and then examined the stomach walls of the mosquitoes, where he found the developing parasites. He then showed that healthy birds bitten by such mosquitoes developed malaria. This proved that mosquitoes are the carriers of malaria, and his work was later confirmed for human malaria in the Anopheles mosquito.

Chapter 2: Biodiversity

A. Multiple Choice Questions

1. Which of the following taxonomic ranks represents the broadest rank?

(a) Species

(b) Genus

(c) Kingdom

(d) Domain ✓

Correct answer: Domain

A. Multiple Choice Questions

2. Which characteristics is unique to organisms in the domain Archaea?

(a) Cell walls made of peptidoglycan

(b) Presence of a nucleus

(c) Ability to live in extreme environments ✓

(d) Lack of ribosomes

Correct answer: Ability to live in extreme environments

A. Multiple Choice Questions

3. Which of the following is a key characteristic that distinguishes eukaryotic cells from prokaryotic cells?

(a) Lack of a cell wall

(b) Presence of a nucleus ✓

(c) Absence of ribosomes

(d) Smaller size

Correct answer: Presence of a nucleus

A. Multiple Choice Questions

4. Why it is impossible to classify viruses within traditional biological kingdoms?

(a) They lack cellular structure and organelles.

(b) They cannot perform photosynthesis.

(c) They are smaller in size than bacteria.

(d) They are parasites.

Correct answer: They lack cellular structure and organelles.

A. Multiple Choice Questions

5. Which of the following is the correct way for writing the scientific name of humans?

(a) Homo sapiens (italic, genus capital, species small)

(b) Homo sapiens (not italic)

(c) Homo Sapiens

(d) homo sapiens

Correct answer: Homo sapiens (italic, genus capital, species small)

B. Short Answers

1. What is the term used to describe the variety of organisms in ecosystems?

The term is biodiversity. It means the variety of all living organisms, including different species, their genetic differences and the ecosystems in which they live.

B. Short Answers

2. How is the biodiversity crucial for humans and for the planet Earth?

Biodiversity keeps ecosystems stable and healthy: plants produce oxygen and food, decomposers recycle nutrients, and many species control pests and pollinate crops. For humans it provides food, medicines, timber and clean air and water. Loss of biodiversity weakens ecosystems and threatens human survival.

B. Short Answers

3. What are the seven taxonomic ranks used in the Linnaean system?

From broadest to narrowest they are: kingdom, phylum, class, order, family, genus and species. In the modern three-domain system, domain stands above kingdom as the broadest rank.

B. Short Answers

4. What are the basic differences between archaea and bacteria?

Both are single-celled prokaryotes without a nucleus, but they differ in their cell walls and chemistry. Archaeal cell walls lack peptidoglycan, which bacterial walls contain, and archaeal cell membranes and genes are different. Archaea often live in extreme environments such as hot springs and salty lakes, while bacteria live almost everywhere.

B. Short Answers

5. Which kingdom includes organisms that are multicellular and heterotrophic, and lack cell walls?

Kingdom Animalia. Animals are multicellular, they obtain food by eating other organisms (heterotrophic nutrition), and their cells have no cell walls.

B. Short Answers

6. Enlist the distinguishing characteristics of fungi.

Fungi are eukaryotic and mostly multicellular, with cell walls made of chitin. They are heterotrophic and absorb food from dead or living organic matter rather than making it. Their bodies are made of thread-like hyphae, and they reproduce by spores.

B. Short Answers

7. List the three domains that encompass all living organisms.

The three domains are Archaea, Bacteria and Eukarya. Archaea and Bacteria contain prokaryotic organisms, while Eukarya contains all organisms with eukaryotic cells: protists, fungi, plants and animals.

B. Short Answers

8. Why cannot we classify viruses in any kingdom?

Viruses are not cells: they have no cell membrane, cytoplasm or organelles, and they cannot carry out life processes on their own. They only reproduce inside a living host cell. Because the kingdom system classifies cellular living organisms, viruses fit into none of its kingdoms.

B. Short Answers

9. How does binomial nomenclature facilitate clear communication about organisms across different languages?

Binomial nomenclature gives every species one unique two-part Latin scientific name, a genus name followed by a species name, for example Homo sapiens. Because the same name is used by scientists in all countries, there is no confusion caused by different local or common names in different languages.

C. Answers in Detail

1. Discuss biodiversity and its significance in maintaining the health of ecosystems.

Biodiversity is the variety of genes, species and ecosystems on Earth. A diverse ecosystem is more stable because different species fill different roles: producers make food, consumers transfer energy, and decomposers return nutrients to the soil. If one species declines, others can partly take its place. Biodiversity also gives humans food, medicines and raw materials, and protects soil, water and

air, so its loss makes ecosystems fragile and less able to recover from change.

C. Answers in Detail

2. Explain the importance of classification in biology and how it helps us understand the relationships between different organisms.

Classification groups organisms by shared characteristics into ranks from domain to species, so that the huge variety of life can be studied in an orderly way. Organisms placed in the same group share a common ancestry and similar features, so classification reveals evolutionary relationships. It also makes identification easy and allows scientists everywhere to communicate about the same organism without confusion.

C. Answers in Detail

3. Compare and contrast the domains Archaea and Bacteria, focusing on their key characteristics.

Both domains contain unicellular prokaryotes with no true nucleus or membrane-bound organelles, and both reproduce mainly by binary fission. They differ in cell wall chemistry: bacterial walls contain peptidoglycan, archaeal walls do not. Their membrane lipids and ribosomal RNA are also different, and archaea are famous for living in extreme conditions such as high temperature, high salt and acidity, while bacteria are found in almost every habitat.

C. Answers in Detail

4. Describe the diagnostic characteristics of the four kingdoms within the domain Eukarya.

Protista contains mostly unicellular eukaryotes, some with plant-like and some with animal-like nutrition. Fungi are heterotrophic eukaryotes with chitin cell walls that absorb food, and their bodies are made of hyphae. Plantae are

multicellular, autotrophic eukaryotes with cellulose cell walls that make food by photosynthesis. Animalia are multicellular, heterotrophic eukaryotes whose cells lack cell walls and which ingest their food.

C. Answers in Detail

5. Discuss the challenges of classifying viruses within the traditional three domains of life.

The three-domain system classifies cellular life, but viruses are acellular particles made only of genetic material in a protein coat. They show no metabolism, growth or independent reproduction, and they are inactive outside a host cell. Some scientists regard them as non-living, others as living only inside a host. Because they share features of both, and have no ribosomes or cell structure to compare, they cannot be placed in any domain or kingdom.

C. Answers in Detail

6. Explain the rules and guidelines for suggesting scientific names to organisms.

As printed in the chapter (book p. 29): every scientific name has two parts, the genus name first followed by the species name. Each name must be unique, so the same name cannot be used for two different organisms. The genus name begins with a capital letter and the species name with a small letter. In print the name is typed in italics, and when handwritten the two parts are underlined separately.

Chapter 3: The Cell

A. Multiple Choice Questions

1. The process of cellular respiration occurs in:

(a) Nucleus

(b) Mitochondria ✓

(c) Ribosomes

(d) Golgi apparatus

Correct answer: Mitochondria

A. Multiple Choice Questions

2. The smooth endoplasmic reticulum (SER) is primarily involved in the synthesis of:

(a) Proteins

(b) Lipids ✓

(c) Carbohydrates

(d) Nucleic acids

Correct answer: Lipids

A. Multiple Choice Questions

3. Which specialized region of the nucleus is responsible for ribosome assembly? (Note: the book prints two options labelled as chromatin; the correct region is the nucleolus.)

(a) Nucleoplasm

(b) Nucleolus ✓

(c) Chromatin

(d) Chromatin

Correct answer: Nucleolus

A. Multiple Choice Questions

4. What is the primary function of ribosomes?

(a) Energy production

(b) Protein synthesis 

(c) Lipid synthesis

(d) DNA synthesis

Correct answer: Protein synthesis

A. Multiple Choice Questions

5. What is the main function of the nuclear pores?

(a) Regulation of cell division

(b) Protein synthesis

(c) Control of pH of the cell

(d) Control of transport of molecules 

Correct answer: Control of transport of molecules

A. Multiple Choice Questions

6. Which cell organelle is responsible for breaking down waste materials?

(a) Golgi apparatus

(b) Nucleus

(c) Mitochondria

(d) Lysosome 

Correct answer: Lysosome

A. Multiple Choice Questions

7. Which of the following cellular structures is found in animal cells and helps in cell division?

(a) Cell membrane

(b) Centriole ✓

(c) Plasmodesma

(d) Vacuole

Correct answer: Centriole

A. Multiple Choice Questions

8. Which sub-cellular organelle plays a crucial role in energy production within the cell?

(a) Endoplasmic reticulum

(b) Golgi apparatus

(c) Mitochondria ✓

(d) Lysosomes

Correct answer: Mitochondria

A. Multiple Choice Questions

9. In a multicellular plant, which cell type is responsible for the production of glucose?

(a) Xylem

(b) Phloem

(c) Epidermal

(d) Mesophyll ✓

Correct answer: Mesophyll

A. Multiple Choice Questions

10. Which organelle can double its number by itself?

(a) Ribosomes

(b) Lysosomes

(c) Mitochondria ✓

(d) Golgi apparatus

Correct answer: Mitochondria

B. Short Answers

1. What are the main functions of cell membrane?

The cell membrane separates the cell from its surroundings and protects its contents. It controls which substances enter and leave the cell, so it is selectively permeable. It also helps the cell communicate with its environment and gives the cell its shape.

B. Short Answers

2. What key role does the Golgi apparatus play in eukaryotic cells?

The Golgi apparatus receives proteins and lipids from the endoplasmic reticulum, modifies them, packs them into vesicles and sends them to their destinations inside or outside the cell. It is therefore the packaging and dispatch centre of the cell.

B. Short Answers

3. How do lysosomes contribute to the cell's functioning?

Lysosomes contain digestive enzymes that break down waste materials, worn-out organelles and foreign particles entering the cell. This cleaning and recycling keeps the cell healthy and supplies it with useful breakdown products.

B. Short Answers

4. Which organelle detoxifies harmful substances and breaks down lipids?

The smooth endoplasmic reticulum (SER). It has no ribosomes on its surface and it synthesises lipids, detoxifies drugs and poisons, and breaks down lipids in the cell.

B. Short Answers

5. What is the smooth endoplasmic reticulum responsible for?

The SER synthesises lipids and steroids, detoxifies harmful substances, and stores and releases calcium ions in muscle cells. It also helps in the breakdown of lipids.

B. Short Answers

6. How do the vacuoles in plant cells differ from vacuoles in animal cells?

A plant cell usually has one large central vacuole that stores water and cell sap and keeps the cell firm. Animal cells have several small vacuoles, if any, used mainly for temporary storage and transport. The plant vacuole is therefore much larger and more permanent.

B. Short Answers

7. What could happen if lysosomal enzymes stop working properly?

Waste materials and worn-out parts would pile up inside the cell instead of being digested. The cell would become damaged and could die, and in the body this leads to storage diseases in which undigested substances accumulate in tissues.

B. Short Answers

8. Why are the cristae important for cellular respiration?

Cristae are the folds of the inner membrane of the mitochondrion. They greatly increase the surface area available for the reactions of aerobic respiration, especially the electron transport chain, so more ATP can be produced.

B. Short Answers

9. How are chromatin and chromosomes related?

Chromatin is the loose, thread-like form of DNA and proteins found in the nucleus

when the cell is not dividing. When the cell prepares to divide, chromatin condenses and coils into thick, visible structures called chromosomes. Chromosomes are therefore the condensed form of chromatin.

B. Short Answers

10. Label the parts of these cell diagrams? (Note: the book's leader lines are fine in print; small repeated bodies such as lysosomes and mitochondria occur more than once, so the same name appears for more than one leader.)

Animal cell (left): 1 Mitochondrion; 2 Cell membrane; 3 Lysosome; 4 Endoplasmic reticulum; 5 Lysosome (small vesicle); 6 Golgi apparatus; 7 Vacuole (small). The large dark body in the centre is the nucleus. Plant cell (right): 1 Golgi apparatus; 2 Nucleus; 3 Endoplasmic reticulum; 4 Central vacuole; 5 Mitochondrion; 6 Mitochondrion (small body); 7 Chloroplast. The thick outer boundary is the cell wall, and the green ovals are chloroplasts.

C. Answers in Detail

1. Explain the fluid mosaic model of the cell membrane.

According to the fluid mosaic model, the cell membrane is a double layer of phospholipids in which proteins are embedded like pieces of a mosaic. The lipid layer is fluid, so the proteins can move sideways within it. Some proteins pass right through the membrane and act as channels or carriers, while others sit on the surface. Cholesterol and carbohydrate chains are also present, and together they give a flexible, selectively permeable boundary.

C. Answers in Detail

2. Describe the structure and functions of the cell wall.

The cell wall is a rigid layer outside the cell membrane, found in plant cells and made mainly of cellulose. It is freely permeable, so water and dissolved

substances pass through it easily. It protects the cell, gives it a definite shape and prevents it from bursting when water enters. In plant cells, neighbouring cells are connected through channels called plasmodesmata that pass through the wall.

C. Answers in Detail

3. Discuss the components of the nucleus.

The nucleus is bounded by a double nuclear membrane containing nuclear pores that control the passage of molecules. Inside is the nucleoplasm, a fluid containing chromatin, the DNA-protein material that carries hereditary information. The nucleolus is a dense region where ribosomes are assembled. The nucleus controls all the activities of the cell and passes genetic information to the next generation.

C. Answers in Detail

4. Describe the structure and function of lysosome and endoplasmic reticulum.

Lysosomes are small round sacs filled with digestive enzymes; they digest food particles, waste and worn-out organelles, acting as the cell's cleaning system. The endoplasmic reticulum is a network of folded membranes: the rough ER carries ribosomes and makes proteins, while the smooth ER makes lipids and detoxifies harmful substances. Both organelles work with the Golgi apparatus to process and transport cell products.

C. Answers in Detail

5. Describe the formation and function of the Golgi complex.

The Golgi complex is formed from stacks of flattened membrane sacs called cisternae, with vesicles budding from their edges. Proteins and lipids arriving from the endoplasmic reticulum enter on one side, are modified, sorted and

packed into vesicles, and leave from the other side. These vesicles carry the products to the cell membrane for secretion or to other parts of the cell, and the Golgi also helps form lysosomes.

C. Answers in Detail

6. Describe the structure and functions of the chloroplast.

The chloroplast is a double-membraned organelle containing a fluid stroma and stacks of disc-like thylakoids called grana. The green pigment chlorophyll sits in the thylakoid membranes and captures light energy. In the thylakoids the light reactions of photosynthesis take place, while the dark reactions take place in the stroma, producing glucose. Chloroplasts are therefore the sites of photosynthesis in plant cells.

Chapter 4: Cell Cycle

A. Multiple Choice Questions

1. In which phase of cell cycle, maximum growth occurs in cell?

(a) M phase

(b) S phase

(c) G1 phase ✓

(d) G2 phase

Correct answer: G1 phase

A. Multiple Choice Questions

2. In which phase of cell cycle, the chromosomes duplicate?

(a) Mitosis

(b) G1 phase

(c) G2 phase

(d) S phase ✓

Correct answer: S phase

A. Multiple Choice Questions

3. Which of the following is NOT a characteristic of mitosis?

(a) It occurs in somatic cells.

(b) It results in genetically identical daughter cells.

(c) The chromosome number is halved in daughter cells. ✓

(d) It results in the formation of two daughter cells.

Correct answer: The chromosome number is halved in daughter cells.

A. Multiple Choice Questions

4. At which stage of mitosis chromosomes line up in the centre?

(a) Prophase

(b) Metaphase ✓

(c) Anaphase

(d) Telophase

Correct answer: Metaphase

A. Multiple Choice Questions

5. If you observe a cell in which nuclear membrane is reforming around two sets of chromosomes, what stage of cell cycle is this?

(a) Anaphase

(b) Telophase ✓

(c) Prophase

(d) Metaphase

Correct answer: Telophase

A. Multiple Choice Questions

6. Why is meiosis-II necessary after meiosis-I?

(a) To replicate chromosomes

(b) To reduce chromosome number

(c) To separate sister chromatids ✓

(d) To ensure genetic recombination

Correct answer: To separate sister chromatids

B. Short Answers

1. Enlist the events that occur during the G1 phase of interphase?

During G1 the cell grows rapidly, makes new proteins and organelles, and carries out its normal functions. It also prepares the materials needed for DNA replication in the next phase. The cell checks that conditions are suitable before it commits to dividing.

B. Short Answers

2. What is the main purpose of the S phase in the cell cycle?

In the S (synthesis) phase the cell copies its DNA, so that each chromosome comes to consist of two identical sister chromatids. This ensures that each daughter cell receives a complete copy of the genetic material after division.

B. Short Answers

3. During which phase of mitosis sister chromatids separate?

Sister chromatids separate during anaphase of mitosis. The spindle fibres pull them apart to opposite poles of the cell, so each pole receives an identical set of chromosomes.

B. Short Answers

4. How does crossing over contribute to genetic variation in meiosis?

During prophase-I of meiosis, homologous chromosomes pair up and exchange segments of DNA, a process called crossing over. This creates new combinations of genes on the chromosomes, so the resulting gametes differ genetically from the parent and from each other.

B. Short Answers

5. What is the role of spindle fibres in mitosis?

Spindle fibres are made of microtubules and attach to the centromeres of chromosomes. During metaphase they line the chromosomes up at the equator, and during anaphase they pull the sister chromatids apart to opposite poles, ensuring an equal distribution of chromosomes.

B. Short Answers

6. How is cytokinesis in animal cell different from plant cell?

In animal cells cytokinesis takes place by a cleavage furrow: the cell membrane pinches inward until the cell divides into two. In plant cells a cell plate forms across the middle of the cell and grows outward to become the new cell wall and membranes, because the rigid wall prevents pinching.

B. Short Answers

7. What is the difference between prophase of mitosis and prophase-I of meiosis-I?

In mitotic prophase, chromosomes condense, the nuclear membrane breaks down and the spindle forms, with no exchange of genetic material. In prophase-I of meiosis, homologous chromosomes pair up (synapsis) and crossing over takes place between them, and prophase-I is much longer, passing through leptotene, zygotene, pachytene, diplotene and diakinesis.

B. Short Answers

8. How does meiosis differ from mitosis in terms of chromosome number?

Mitosis produces two daughter cells with the same chromosome number as the parent (diploid stays diploid). Meiosis produces four daughter cells with half the chromosome number (diploid becomes haploid), because the chromosome number is halved in meiosis-I and the chromatids are then separated in meiosis-II.

B. Short Answers

9. What are the key events of anaphase in mitosis?

In anaphase the centromeres divide and the sister chromatids separate. Spindle fibres shorten and pull the chromatids, now called daughter chromosomes, to opposite poles of the cell. By the end of anaphase each pole has an identical set of chromosomes.

B. Short Answers

10. What is the function of the centrosome during cell division?

The centrosome is the microtubule-organising centre of the cell. During division it duplicates and its two copies move to opposite poles, where they organise the spindle fibres that move the chromosomes. In animal cells each centrosome contains a pair of centrioles.

B. Short Answers

11. What are sister chromatids, and when do they separate in meiosis?

Sister chromatids are the two identical copies of a chromosome formed when DNA replicates in the S phase, joined at the centromere. In meiosis they stay together through meiosis-I and finally separate during anaphase-II of meiosis-II, so each gamete receives one chromatid.

B. Short Answers

12. How is mitosis related to the process of regeneration?

Regeneration is the regrowth of lost or damaged body parts, and it depends on mitosis. Mitosis produces new, genetically identical cells that replace the lost cells and rebuild the tissue, as in the healing of a wound or the regrowth of a lizard's tail.

C. Answers in Detail

1. Describe the events that occur during the phases of mitosis.

In prophase, chromosomes condense and become visible, the nuclear membrane breaks down and the spindle begins to form. In metaphase, chromosomes line up at the equator of the cell, attached to spindle fibres. In anaphase, sister chromatids separate and move to opposite poles. In telophase, nuclear membranes reform around the two sets of chromosomes and the chromosomes

uncoil. Cytokinesis then divides the cytoplasm into two daughter cells.

C. Answers in Detail

2. Describe cytokinesis in animal and plant cells.

In animal cells, a ring of microfilaments beneath the membrane contracts and pinches the cell inward, forming a cleavage furrow that deepens until two daughter cells are produced. In plant cells, vesicles from the Golgi apparatus gather at the equator and fuse to form a cell plate, which grows outward until it joins the parent walls and becomes the new cell wall and membranes separating the two daughter cells.

C. Answers in Detail

3. Describe the significance of mitosis.

Mitosis is essential for growth, because it increases the number of cells in a multicellular organism. It replaces worn-out and damaged cells, heals wounds and allows regeneration of lost parts. Because the daughter cells are genetically identical to the parent, mitosis maintains the chromosome number and passes the same hereditary information to every body cell.

C. Answers in Detail

4. Describe the events that occur during the phases of meiosis-I.

In prophase-I, homologous chromosomes pair up and crossing over exchanges segments between them. In metaphase-I, the homologous pairs line up at the equator, with one homologue facing each pole. In anaphase-I, the homologous chromosomes separate and move to opposite poles, while sister chromatids stay together. In telophase-I, the cell divides into two cells, each with half the chromosome number, and the chromosomes may partially uncoil.

C. Answers in Detail

5. Describe the significance of meiosis.

Meiosis halves the chromosome number, so that when two gametes fuse at fertilisation the diploid number is restored. It produces genetic variation through crossing over and the independent assortment of chromosomes, which is the raw material for evolution. Meiosis also produces the gametes needed for sexual reproduction, giving offspring new combinations of characters.

Chapter 5: Tissues, Organs, and Organ Systems

A. Multiple Choice Questions

1. At which level of organization gas exchange occurs between body and environment? (Note: exchange across the alveolar surface itself is cellular, but the body-environment exchange is carried out by the respiratory system as a whole.)

(a) Organelle level in mitochondria

(b) Cellular level in alveolar cells

(c) Tissue level in epithelial tissues

(d) Organ system level in the respiratory system ✓

Correct answer: Organ system level in the respiratory system

A. Multiple Choice Questions

2. The epithelial tissue in the stomach wall is responsible for producing:

(a) Mucus

(b) Pepsinogen

(c) Hydrochloric acid

(d) All of these ✓

Correct answer: All of these

A. Multiple Choice Questions

3. In the wall of stomach, which tissue also contains blood vessels and nerves?

(a) Epithelial

(b) Muscle

(c) Inner connective ✓

(d) Outer connective

Correct answer: Inner connective

A. Multiple Choice Questions

4. In a leaf, which tissue is responsible for photosynthesis?

(a) Xylem

(b) Mesophyll ✓

(c) Epidermis

(d) Phloem

Correct answer: Mesophyll

A. Multiple Choice Questions

5. What is the primary function of the xylem tissue in a leaf?

(a) To transport sugars to other parts

(b) To transport water to parts of leaf ✓

(c) To synthesize chlorophyll

(d) To control the opening and closing of stomata

Correct answer: To transport water to parts of leaf

B. Short Answers

1. Enlist the levels of organization from cells to organ systems.

The levels are: cell, tissue, organ, organ system and organism. Cells of the same type form a tissue, different tissues form an organ, organs working together form an organ system, and all the systems together make the organism.

B. Short Answers

2. What are the major roles of the epithelial tissue present in the stomach?

The epithelial tissue lines the inner wall of the stomach and protects it from the acid and enzymes of the gastric juice. Its cells secrete mucus, which coats and protects the wall, and they also produce pepsinogen and hydrochloric acid, which begin the digestion of proteins.

B. Short Answers

3. How do the smooth muscles contribute to the stomach's function?

The stomach wall has three layers of smooth muscle: outer longitudinal, middle circular and inner oblique. These muscles contract and relax in turn, churning the food and mixing it thoroughly with gastric juice, and they push the partially digested food onward into the small intestine.

B. Short Answers

4. What is the function of the palisade mesophyll in the leaf?

The palisade mesophyll is the upper layer of mesophyll tissue, made of tall, closely packed cells rich in chloroplasts. Because it lies just below the upper epidermis, it receives the most light and is the main site of photosynthesis in the leaf.

B. Short Answers

5. Differentiate between the following: i. Tissue and organ; ii. Root system and shoot system; iii. Epidermal and mesophyll tissue; iv. Palisade and spongy mesophyll.

i. A tissue is a group of similar cells performing one function, while an organ is made of several tissues working together for a larger function. ii. The root system grows underground and absorbs water and salts, while the shoot system grows above ground and carries out photosynthesis and reproduction. iii. Epidermal tissue is the protective outer layer of the leaf, while mesophyll tissue lies inside and carries out photosynthesis. iv. Palisade mesophyll consists of tall, tightly packed cells with many chloroplasts, while spongy mesophyll consists of loosely arranged cells with air spaces, which allow gas exchange.

C. Answers in Detail

1. Explain the levels of organization in multicellular organism. How does each level contribute to the overall functioning of an organism?

The cell is the basic unit of life and carries out all life processes. Similar cells form a tissue, such as muscle tissue, which performs one job better than a single cell could. Different tissues combine into an organ, such as the stomach, which performs a complete function. Organs working together form an organ system, such as the digestive system, and all the systems together make the organism. Each level adds organisation, so the whole organism can carry out complex functions efficiently.

C. Answers in Detail

2. What is a tissue level? Explain plant and animal tissues.

The tissue level is the level at which groups of similar cells work together for a common function. In plants the main tissues are meristematic tissue, which divides and produces new cells, and permanent tissues such as epidermis, parenchyma, xylem and phloem. In animals the four main tissues are epithelial tissue (covering and lining), connective tissue (support and binding), muscle tissue (movement) and nervous tissue (control and coordination).

C. Answers in Detail

3. Describe the tissue composition of the stomach. How does each tissue contribute to the digestive function of the stomach?

The stomach wall has four layers. The innermost epithelial tissue secretes mucus, pepsinogen and hydrochloric acid, protecting the wall and starting protein digestion. Beneath it, connective tissue supports the wall and carries blood vessels, nerves and lymphatics. Three layers of smooth muscle churn the food and mix it with gastric juice. The outermost connective tissue encircles and supports the whole organ. Together these tissues let the stomach store, churn

and chemically digest food.

C. Answers in Detail

4. Describe the tissue composition of the leaf. How does each tissue contribute to the functions of the leaf?

The leaf is covered above and below by epidermal tissue, which protects it and controls water loss through stomata. Inside lies the mesophyll: the palisade layer, rich in chloroplasts, is the main photosynthetic tissue, while the spongy layer has air spaces for gas exchange. Vascular tissue runs through the veins: xylem brings water to the leaf cells, and phloem carries away the sugars made in photosynthesis. Together these tissues let the leaf make food, exchange gases and transport materials.

Chapter 6: Enzymes

A. Multiple Choice Questions

1. Primarily, all enzymes are;

(a) Nucleic acids

(b) Proteins ✓

(c) Carbohydrates

(d) Lipids

Correct answer: Proteins

A. Multiple Choice Questions

2. Which best defines an enzyme?

(a) A chemical that breaks down food.

(b) A hormone that regulates metabolism.

(c) A protein that speeds up reactions. ✓

(d) A molecule that stores energy.

Correct answer: A protein that speeds up reactions.

A. Multiple Choice Questions

3. Enzymes are specific in their action because:

(a) Their active sites fit specific substrates. ✓

(b) They are always proteins.

(c) They are consumed in reactions.

(d) They work only at high temperatures.

Correct answer: Their active sites fit specific substrates.

A. Multiple Choice Questions

4. Prosthetic groups are;

(a) Required by all enzymes.

(b) Proteins in nature.

(c) Loosely attached with enzymes.

(d) Tightly bound to enzyme. ✓

Correct answer: Tightly bound to enzyme.

A. Multiple Choice Questions

5. What is TRUE according to the induced fit model of enzyme action?

(a) Enzyme's active site changes shape to bind the substrate. ✓

(b) Substrate changes shape to bind to active site.

(c) No shape changes occur in active site or substrate.

(d) Substrate attaches the enzyme at a site other that active site.

Correct answer: Enzyme's active site changes shape to bind the substrate.

B. Short Answers

1. Define metabolism. Differentiate between catabolism and anabolism.

Metabolism is the sum of all chemical reactions taking place in a living cell or body. Catabolism is the breakdown of complex molecules into simpler ones, releasing energy, for example the breakdown of glucose in respiration. Anabolism is the building up of complex molecules from simpler ones, which needs energy, for example the synthesis of proteins from amino acids.

B. Short Answers

2. Which type of metabolism demands input of energy? Give an example.

Anabolism demands an input of energy, because small molecules are joined together to build larger ones. An example is photosynthesis, in which light energy is used to build glucose from carbon dioxide and water, or the synthesis of proteins from amino acids.

B. Short Answers

3. Define an enzyme. What is its role in metabolism?

An enzyme is a protein that acts as a biological catalyst: it speeds up a chemical reaction without being used up in it. In metabolism, enzymes lower the activation energy of reactions, so that the breakdown and building reactions of the cell take place fast enough to support life at body temperature.

B. Short Answers

4. What is the active site of enzyme? State its importance in enzyme specificity.

The active site is a pocket or cleft on the surface of the enzyme into which the substrate fits. Its shape and chemical nature match only a particular substrate, just as a lock fits one key, and this is why each enzyme acts on only one substrate or a small group of related substrates.

B. Short Answers

5. Provide an example of a specific enzyme-substrate pair.

Amylase and starch: amylase acts only on starch, breaking it down into maltose, and does not act on proteins or fats. Another example is pepsin, which acts on proteins in the stomach, and lipase, which acts on lipids.

C. Answers in Detail

1. Describe the characteristics of enzymes.

Enzymes are proteins that act as biological catalysts and are not used up in the reaction, so a small amount can catalyse many reactions. They are highly specific, each acting on a particular substrate because of the shape of its active site. They work best at an optimum temperature and pH, and are destroyed by extreme heat or pH. They lower the activation energy of a reaction and speed it up, but do not change the products formed.

C. Answers in Detail

2. Describe the factors that affect the activity of enzymes.

Temperature: activity rises with temperature up to an optimum (around body temperature in humans) and then falls sharply as the enzyme denatures. pH: each enzyme has an optimum pH, and strong acids or alkalis denature it. Substrate concentration: activity rises as more substrate is available, until all active sites are occupied. Enzyme concentration, and the presence of activators, coenzymes or inhibitors, also change the rate of reaction.

C. Answers in Detail

3. Compare the Lock-and-Key and Induced Fit models of enzyme action.

In the Lock-and-Key model, proposed by Emil Fischer in 1894, the active site has a fixed, rigid shape and the substrate fits into it exactly like a key into a lock. In the Induced Fit model, proposed by Daniel Koshland in 1958, the active site is flexible: when the substrate enters, the active site changes shape slightly to fit it more closely, and this strain helps the reaction. The induced fit model explains why enzymes can act on slightly different substrates, which the rigid lock-and-key model cannot.

Chapter 7: Bioenergetics

A. Multiple Choice Questions

1. When we get energy from ATP, which bonds are broken?

(a) P-P bonds ✓

(b) C-H bonds

(c) C-N bonds

(d) C-O bonds

Correct answer: P-P bonds

A. Multiple Choice Questions

2. Light reactions of photosynthesis occur in;

(a) Plasma membrane of cell

(b) Cytoplasm of cell

(c) Stroma of chloroplasts

(d) Thylakoids of chloroplasts ✓

Correct answer: Thylakoids of chloroplasts

A. Multiple Choice Questions

3. Which wavelengths of light are absorbed to maximum by chlorophylls?

(a) Green and blue

(b) Green and red

(c) Red and blue ✓

(d) Only green

Correct answer: Red and blue

A. Multiple Choice Questions

4. Where do the dark reactions of photosynthesis occur?

(a) Stroma of chloroplast ✓

(b) Thylakoids of chloroplast

(c) Outer membrane

(d) Cytoplasm

Correct answer: Stroma of chloroplast

A. Multiple Choice Questions

5. Which process in aerobic respiration produces the most ATP?

(a) Glycolysis

(b) Krebs cycle

(c) Electron transport chain

(d) Fermentation

Correct answer: Electron transport chain

B. Short Answers

1. Write the importance of oxidation-reduction reactions.

Oxidation-reduction (redox) reactions transfer energy in living cells. When a molecule is oxidised it loses electrons and energy is released, and when another is reduced it gains those electrons. Photosynthesis and respiration are both redox processes, and they drive the formation and use of ATP, the cell's energy currency.

B. Short Answers

2. What do ATP and ADP mean? What are the roles of these molecules for the cellular metabolism?

ATP is adenosine triphosphate and ADP is adenosine diphosphate. ATP stores energy in the bonds between its phosphate groups; when the last phosphate is removed, ATP becomes ADP and energy is released for cell work such as muscle contraction and active transport. ADP is then recharged to ATP using energy

from respiration, so the ATP-ADP cycle powers metabolism.

B. Short Answers

3. Write down the word equation for photosynthesis.

Carbon dioxide + water, in the presence of light energy and chlorophyll, gives glucose + oxygen. In symbols: 6CO2 + 6H2O gives C6H12O6 + 6O2.

B. Short Answers

4. How is oxygen produced during photosynthesis?

During the light reactions in the thylakoids, light energy splits water molecules, a process called photolysis. Water splits into hydrogen ions, electrons and oxygen gas. The oxygen is released into the air through the stomata, while the hydrogen and electrons are used to make ATP and NADPH for the dark reactions.

B. Short Answers

5. Which organisms carry out photosynthesis? Which cell organelle is responsible for the absorption of light for photosynthesis?

Photosynthesis is carried out by green plants, algae and some bacteria such as cyanobacteria. In plant cells the organelle responsible for absorbing light is the chloroplast, whose pigment chlorophyll in the thylakoid membranes captures light energy.

B. Short Answers

6. State the main purpose of cellular respiration?

The main purpose of cellular respiration is to release the energy stored in food, especially glucose, and to transfer it into ATP. ATP then supplies energy for all the activities of the cell, such as movement, growth, active transport and the synthesis of new molecules.

B. Short Answers

7. State the equation (in words or symbols) for aerobic respiration.

Glucose + oxygen gives carbon dioxide + water + energy (ATP). In symbols: C6H12O6 + 6O2 gives 6CO2 + 6H2O + energy. The book's Figure 7.8 gives the total as 36 ATP from one glucose molecule in aerobic respiration.

B. Short Answers

8. Write a brief note on the role of oxygen in aerobic respiration.

Oxygen acts as the final acceptor of electrons and hydrogen at the end of the electron transport chain. It combines with them to form water, which keeps the chain running and allows large amounts of ATP to be made. Without oxygen, the chain stops and the cell must rely on anaerobic respiration, which yields far less energy.

B. Short Answers

9. Define anaerobic and aerobic respiration.

Aerobic respiration is the breakdown of glucose in the presence of oxygen, producing carbon dioxide, water and a large amount of energy (36 ATP per glucose in this book). Anaerobic respiration is the breakdown of glucose without oxygen, producing much less energy, and in animals lactic acid, or in yeast ethanol and carbon dioxide, as by-products.

C. Answers in Detail

1. Explain ATP as a molecule that is the chief energy currency of all cells.

ATP (adenosine triphosphate) consists of adenine, ribose sugar and three phosphate groups. Energy is stored in the bonds between the phosphate groups, especially the last one. When a cell needs energy, ATP is hydrolysed to ADP (adenosine diphosphate) and a phosphate, releasing energy for work such as

contraction, transport and synthesis. Because every cell makes ATP in respiration and spends it in the same way, ATP is called the energy currency of the cell.

C. Answers in Detail

2. Outline the processes involved in photosynthesis?

Photosynthesis has two stages. In the light reactions, which take place in the thylakoids, chlorophyll absorbs light, water is split to release oxygen, and ATP and NADPH are formed. In the dark reactions (Calvin cycle), which take place in the stroma, carbon dioxide is fixed using the ATP and NADPH to build glucose. Light energy is thus converted into chemical energy stored in food.

C. Answers in Detail

3. Write a note on the intake of carbon dioxide and water by plants.

Carbon dioxide enters the leaf from the air through the stomata, the small pores in the epidermis controlled by guard cells. Water is absorbed from the soil by root hairs and carried up through the xylem to the leaves. In the leaf, carbon dioxide and water meet in the chloroplasts, where they are used in photosynthesis to make glucose and release oxygen.

C. Answers in Detail

4. Explain the types and importance of anaerobic respiration.

The two main types are lactic acid fermentation in animal muscle cells, which produces lactic acid and a little energy, and alcoholic fermentation in yeast and some bacteria, which produces ethanol and carbon dioxide. Anaerobic respiration is important because it lets cells make some ATP when oxygen is absent, for example during vigorous exercise, and it is used in industry for baking and brewing. Its drawback is the small energy yield and the harmful by-products.

C. Answers in Detail

5. Outline the mechanism of aerobic respiration.

Aerobic respiration has three main stages. In glycolysis in the cytoplasm, glucose is split into two molecules of pyruvic acid, giving a net gain of 2 ATP. Pyruvic acid enters the mitochondria, is converted to acetyl CoA, and passes through the Krebs cycle, releasing carbon dioxide and hydrogen carriers (NADH and FADH2). In the electron transport chain on the inner mitochondrial membrane, these carriers release their energy to make most of the ATP, and oxygen accepts the final electrons to form water, giving 36 ATP in total in this book's account.

Chapter 8: Plant Physiology

A. Multiple Choice Questions

1. The primary function of root hairs is:

(a) Transport of nutrients

(b) Storage of food

(c) Increase surface area for absorption ✓

(d) Synthesis of proteins

Correct answer: Increase surface area for absorption

A. Multiple Choice Questions

2. Water moves from the soil into root cells by:

(a) Osmosis ✓

(b) Active transport

(c) Diffusion

(d) Bulk flow

Correct answer: Osmosis

A. Multiple Choice Questions

3. The transpiration is regulated by;

(a) Mesophyll

(b) Guard cells ✓

(c) Xylem

(d) Phloem

Correct answer: Guard cells

A. Multiple Choice Questions

4. In most plants the food is transported in the form of:

(a) Glucose

(b) Sucrose

(c) Starch

(d) Maltose

Correct answer: Sucrose

A. Multiple Choice Questions

5. What is TRUE according to the pressure flow mechanism of food transport?

(a) Water enters the source, creating pressure

(b) Water is pulled from the sink

(c) Movement of food in phloem is due to gravity

(d) Solutes move from low to high concentration

Correct answer: Water enters the source, creating pressure

A. Multiple Choice Questions

6. Succulent organs are present in:

(a) Xerophytes

(b) Hydrophytes

(c) Mesophytes

(d) Halophytes

Correct answer: Xerophytes

B. Short Answers

1. Define transpiration and its types.

Transpiration is the loss of water vapour from the aerial parts of a plant, mainly through the leaves. Its types are stomatal transpiration (through stomata, the main type), cuticular transpiration (through the cuticle of the leaf surface) and lenticular transpiration (through lenticels in stems).

B. Short Answers

2. How is the transpirational pull important in plants?

As water evaporates from the leaves, a pull (tension) is created that draws water upward through the xylem from the roots. This transpirational pull is the main force lifting water and dissolved salts to the top of tall plants, without any expenditure of energy by the plant.

B. Short Answers

3. Transpiration is the loss of water from plants. Is it a harmful phenomenon? If no, what is its importance?

No, transpiration is not harmful; it is a necessary evil with important benefits. It pulls water and minerals up from the roots, cools the leaves, and helps in the distribution of salts through the plant. Although it loses water, the plant gains far more from the upward transport and cooling it provides.

B. Short Answers

4. Differentiate between: i. Xylem and phloem; ii. Transpiration and guttation; iii. Hydrophytes and halophytes; iv. Hydrophytes and xerophytes; v. Lenticular transpiration and stomatal transpiration.

i. Xylem transports water and salts upward and is made of dead cells, while phloem transports food in both directions and is made of living cells. ii. Transpiration is the loss of water as vapour, while guttation is the loss of liquid water as droplets from leaf margins. iii. Hydrophytes live in water, while halophytes live in salty habitats and tolerate salt. iv. Hydrophytes live in water with abundant supply, while xerophytes live in dry deserts and conserve water. v. Lenticular transpiration takes place through lenticels in stems, while stomatal transpiration takes place through stomata in leaves.

C. Answers in Detail

1. Describe the events involved in the opening and closing of stomata.

Stomata are pores bounded by two guard cells. When guard cells take in water by osmosis they become turgid and bow outward, opening the pore. When they lose water they become flaccid and close together, shutting the pore. Light, potassium ion movement and the water supply of the leaf control this turgor change, so stomata open in daylight for photosynthesis and gas exchange, and close to reduce water loss.

C. Answers in Detail

2. Explain the internal structure of root and describe the uptake of salt and water by root.

The root in cross-section has an outer epidermis with root hairs, a cortex of parenchyma cells, an endodermis, and a central vascular cylinder containing xylem and phloem. Water enters root hairs by osmosis, because the cell sap is more concentrated than the soil solution, and passes through the cortex to the xylem. Salts are taken up mainly by active transport, using energy, and then move with the water into the xylem for transport upward.

C. Answers in Detail

3. Describe the mechanism of transport of water and salt in plants.

Water is absorbed by root hairs by osmosis and moves through the cortex into the xylem vessels. It then rises through the xylem by the combined forces of root pressure, capillary action, cohesion between water molecules and adhesion to vessel walls, and above all the transpirational pull from the leaves. Salts move with the water, having been taken up by the roots by active transport, and are distributed to all parts of the plant.

C. Answers in Detail

4. Explain the mechanism of food translocation by Pressure Flow Mechanism.

In the pressure flow mechanism, sugars made in the leaves (the source) are actively loaded into the phloem sieve tubes. Water follows by osmosis, raising the pressure at the source end. At the sink (roots, fruits or growing parts), sugars are removed for use or storage, water leaves the phloem, and the pressure falls. This pressure difference pushes the sugary sap from source to sink through the phloem.

C. Answers in Detail

5. Explain osmotic adjustments in hydrophytes, xerophytes and halophytes.

Hydrophytes live in water, so they face no shortage: they have broad floating leaves with many stomata on the upper surface, and water moves out freely (the water lily is the book's example). Xerophytes live in deserts: they have deep roots, very few stomata and a thick waxy cuticle to cut water loss, and some, like cacti, store water in soft, juicy succulent stems or roots. Halophytes live in salty water: water tends to leave their bodies, so they absorb salts to make their cells hypertonic, and excrete excess salt through salt glands on their leaves; many sea grasses belong to this group.

Chapter 9: Reproduction in Plants

A. Multiple Choice Questions

1. Which of the following organisms commonly reproduce by binary fission?

(a) Yeast

(b) Bacteria ✓

(c) Rhizopus

(d) Plants

Correct answer: Bacteria

A. Multiple Choice Questions

2. What is the primary method of reproduction in yeast?

(a) Binary fission

(b) Spore formation

(c) Budding ✓

(d) Fragmentation

Correct answer: Budding

A. Multiple Choice Questions

3. What happens in some bacteria during harsh conditions?

(a) Creation of a bud that detaches from the cell

(b) Formation of thick-walled endospores ✓

(c) Splitting the cell into two identical daughter cells

(d) Fusion of two bacterial cells

Correct answer: Formation of thick-walled endospores

A. Multiple Choice Questions

4. Which of the following is an example of vegetative propagation through runners?

(a) Potato

(b) Strawberry ✓

(c) Onion

(d) Ginger

Correct answer: Strawberry

A. Multiple Choice Questions

5. Which plant propagates through tubers?

(a) Onion

(b) Potato ✓

(c) Ginger

(d) Garlic

Correct answer: Potato

A. Multiple Choice Questions

6. Which of these does NOT help a plant for vegetative propagation?

(a) Rhizome

(b) Corm

(c) Runner

(d) Flower ✓

Correct answer: Flower

A. Multiple Choice Questions

7. Which of the following is NOT a part of carpel?

(a) Filament ✓

(b) Style

(c) Stigma

(d) Ovary

Correct answer: Filament

A. Multiple Choice Questions

8. Which structure forms the female gametophyte in flowering plants?

(a) Pollen grain

(b) Ovule ✓

(c) Anther

(d) Sepal

Correct answer: Ovule

A. Multiple Choice Questions

9. The male gametophyte in flowering plants is known as:

(a) Pollen grain ✓

(b) Embryo sac

(c) Ovary

(d) Carpel

Correct answer: Pollen grain

A. Multiple Choice Questions

10. In the life cycle of flowering plants, which structure is triploid (3n)?

(a) Egg

(b) Fusion nucleus

(c) Endosperm nucleus ✓

(d) Sperm

Correct answer: Endosperm nucleus

A. Multiple Choice Questions

11. Double fertilization involves;

(a) Fertilization of the egg by two male gametes

(b) Fertilization of two eggs in the same embryo sac by two sperms

(c) Fertilization of the egg and the fusion nucleus by two sperms ✓

(d) Fertilization of the egg and the tube cell by two sperms

Correct answer: Fertilization of the egg and the fusion nucleus by two sperms

B. Short Answers

1. Write a short note on budding in yeast.

In budding, a small outgrowth (bud) appears on the yeast cell and grows, while the nucleus divides and one daughter nucleus moves into the bud. The bud then separates from the parent cell and grows into a new yeast cell. Sometimes buds remain attached for a while, forming a chain of cells.

B. Short Answers

2. Write a short note on spore formation in fungi.

In fungi such as Rhizopus, spores are produced in sac-like sporangia at the tips of aerial hyphae. When the sporangium bursts, the light spores are scattered by wind. Each spore that lands in suitable conditions germinates and grows into a new fungal body.

B. Short Answers

3. What are the advantages of spore formation in fungi and bacteria?

Spores are light and easily dispersed by wind and water, so the organism spreads to new places. In bacteria, thick-walled endospores survive harsh conditions such as heat, dryness and lack of food, and germinate when conditions improve. Spore formation therefore aids both dispersal and survival.

B. Short Answers

4. Describe how vegetative propagation occurs through runners.

Runners (stolons) are horizontal stems that grow along the surface of the ground from the parent plant. At their nodes they produce roots and new shoots, which grow into independent plants. Strawberry is the common example: one parent plant can thus produce many new plants around itself.

B. Short Answers

5. State how potatoes reproduce through tubers.

A potato tuber is an underground stem swollen with stored food, bearing buds in its eyes. When a tuber, or a piece of it with an eye, is planted, the buds sprout into new shoots and roots, and a new potato plant grows. The new plant is genetically identical to the parent.

B. Short Answers

6. Describe the advantages and disadvantages of vegetative propagation.

Advantages: new plants are identical to the parent, so good characters are preserved; it is fast, and plants that do not produce seeds can still be multiplied.

Disadvantages: there is no genetic variation, so all the plants are equally vulnerable to disease and changing conditions, and diseases of the parent are passed on to the offspring.

B. Short Answers

7. Name the four whorls present in a flower and also tell the components of each whorl.

The four whorls are: calyx, made of sepals, which protect the bud; corolla, made of petals, which attract pollinators; androecium, the male whorl, made of stamens, each with a filament and an anther; and gynoecium, the female whorl, made of carpels, each with a stigma, style and ovary.

B. Short Answers

8. Briefly describe the formation of egg cell and polar nuclei within embryo sac of a flower.

Inside the ovule, a megaspore mother cell divides by meiosis to form four megaspores, of which one survives. Its nucleus divides three times by mitosis to form eight nuclei in the embryo sac. These organise into an egg cell with two synergids at one end, three antipodal cells at the other, and two polar nuclei in the centre, which fuse to form the diploid fusion nucleus.

B. Short Answers

9. Differentiate between: i. Asexual and sexual reproduction; ii. Binary fission in bacteria and amoeba; iii. Stolon and rhizome; iv. Bulb and corm; v. Cutting and grafting; vi. Vegetative propagation and artificial propagation; vii. Male and female gametophytes; viii. Calyx and corolla; ix. Stamen and carpel.

i. Asexual reproduction uses one parent and produces identical offspring without gametes, while sexual reproduction uses two parents and gametes, producing varied offspring. ii. In bacteria, binary fission is division of a single cell into two identical cells, while in Amoeba the same process is a form of asexual reproduction in a unicellular animal. iii. A stolon (runner) grows above the ground, while a rhizome is an underground horizontal stem. iv. A bulb is a short stem with fleshy leaves (onion), while a corm is a solid swollen stem base (gladiolus). v. In cutting, a piece of stem or leaf is planted to grow a new plant, while in grafting a scion is joined onto a rootstock. vi. Vegetative propagation is natural reproduction from plant parts, while artificial propagation is the same process carried out deliberately by humans. vii. The male gametophyte is the pollen grain, while the female gametophyte is the embryo sac. viii. The calyx is the whorl of sepals, while the corolla is the whorl of petals. ix. The stamen is the male part (filament + anther), while the carpel is the female part (stigma + style

+ ovary).

B. Short Answers

10. Label the given diagram of flower. (Note: the book's flower diagram carries leader lines without printed numbers, so the parts are listed in order from the top of the flower downward.)

From the top downward, the labelled parts are: petal; anther (the pollen-bearing head of the stamen); filament (the stalk of the stamen); stigma (the receptive tip of the carpel); style (the stalk of the carpel); ovary (the swollen base of the carpel containing ovules); ovule (inside the ovary); sepal (the green leaf-like part below the petals); receptacle (the base on which the flower parts sit); and pedicel (the stalk of the flower).

C. Answers in Detail

1. Explain the process of binary fission in bacteria and describe how it leads to the formation of two daughter bacteria.

In binary fission, the single circular DNA of the bacterium first replicates, and the two copies move to opposite ends of the cell. A cross-wall (septum) then grows inward across the middle of the cell, and the cell membrane and wall divide. The result is two genetically identical daughter bacteria, each with its own copy of DNA, and under favourable conditions this can repeat every 20 minutes or so.

C. Answers in Detail

2. What do you mean by vegetative propagation? Differentiate among different plant structures modified for vegetative propagation.

Vegetative propagation is asexual reproduction in which a new plant grows from a vegetative part of the parent, such as a stem, root or leaf, without seeds. Modified stems include runners (strawberry, above ground), rhizomes (ginger,

underground), tubers (potato, swollen underground stem), bulbs (onion, fleshy leaves on a short stem) and corms (solid underground stem bases). Roots and leaves can also produce new plants in some species, and all the offspring are genetically identical to the parent.

C. Answers in Detail

3. Describe the ways by which humans can grow new plants by using the vegetative parts of the parent plants?

Humans use cuttings, in which a piece of stem, root or leaf is cut and planted until it roots. In layering, a branch is bent to the ground and covered with soil until it roots, then separated. In grafting, a scion (shoot) of a desired variety is joined to the rootstock of another plant so they grow as one. Tissue culture grows many plantlets from a small piece of tissue in a laboratory medium. All these methods preserve the parent's characters exactly.

C. Answers in Detail

4. Define sporophyte and gametophyte. State their roles in the life cycle of plants.

The sporophyte is the diploid (2n) generation that produces spores by meiosis in sporangia. The gametophyte is the haploid (n) generation that grows from a spore and produces gametes by mitosis. In the plant life cycle these two generations alternate: the sporophyte gives rise to the gametophyte through spores, and the gametophyte gives rise to the next sporophyte when gametes fuse at fertilisation.

C. Answers in Detail

5. Explain the lifecycle of flowering plants, focusing on the alternation between the gametophyte and sporophyte generations.

The flowering plant itself is the diploid sporophyte. In its flowers, meiosis produces haploid spores: microspores in the anthers and megaspores in the ovules. These grow into the gametophytes: the pollen grain (male) and the embryo sac (female). The pollen grain produces two male gametes, and the embryo sac contains the egg. At double fertilisation, one sperm fuses with the egg to form the diploid zygote, which grows into the new sporophyte embryo, while the other fuses with the fusion nucleus to form the triploid endosperm that feeds it.

C. Answers in Detail

6. Describe how the female gametophyte (embryo sac) develops within the ovule of a flower.

Within the ovule, a diploid megaspore mother cell divides by meiosis into four haploid megaspores, three of which degenerate and one survives. The surviving megaspore's nucleus divides three times by mitosis, producing eight nuclei. These organise into the seven-celled embryo sac: an egg cell and two synergids at the micropylar end, three antipodal cells at the opposite end, and two polar nuclei in the centre. This embryo sac is the female gametophyte, ready for fertilisation.